

DED-SAM: Adapting Segment Anything Model 2 for Dual Encoder–Decoder Change Detection

Junlong Qiu , Wei Liu , Xin Zhang , Erzhu Li , Lianpeng Zhang , and Xing Li 

Abstract—Change detection has become a crucial topic in the field of remote sensing deep learning due to its extensive application in earth observation. However, real remote sensing images often contain multiple land cover classes with significant intra-class variability and interclass similarity, limiting the performance of change detection in complex scenarios. To address this, we leverage the capabilities of vision foundation models by applying the segment anything model (SAM) to remote sensing change detection, and we name this method dual encoder–decoder SAM (DED-SAM). Specifically, we construct a DED framework, utilizing a small-scale change detection model in both branches to generate mixed prompts including image features, mask prompts, and box prompts. The SAM 2 model is used for fine-grained recognition of dual-temporal images, generating accurate, and stable feature boundaries, which are then used as constraints to generate the final change mask. To validate the effectiveness of DED-SAM across various application scenarios, we conduct quantitative experiments on three public datasets: Levir-CD, SYSU-CD, and CDD, testing its detection capabilities under single change categories, multiple change categories, and seasonal pseudochange interference. The results show that the proposed DED-SAM achieved state-of-the-art F1 scores and IoUs on these three datasets: LEVIR-CD (92.00%, 85.11%), SYSU-CD (84.15%, 72.01%), and CDD (97.72%, 95.47%).

Index Terms—Change detection, remote sensing, segment anything model, vision foundation model.

I. INTRODUCTION

REMOTE sensing change detection is a technique that relies on multitemporal images of the same area to detect and locate changes within that region [1]. In recent years, advancements in Earth observation technology have made it possible to acquire high-resolution, large-scale, and frequently updated

remote sensing imagery, which has further raised the requirements for accuracy and efficiency in change detection. Deep learning techniques provide more stable and accurate decisions by extracting features from vast amounts of data. Utilizing deep learning methods for remote sensing image processing can effectively leverage the massive data made available by advancements in remote sensing technology, achieving superior efficiency, and performance compared to traditional methods. This has led to widespread applications in various fields, including change detection, land use surveys, environmental monitoring, and disaster assessment [2], [3], [4]. Change detection methods based on deep learning can be categorized into object-oriented [5], [6], [7] and pixel-based methods [8], [9].

Object-oriented change detection methods typically use image patches as the basic input units, comparing classification results from dual-temporal images to identify change areas. The detection accuracy of these methods often depends on the segmentation of input images and postprocessing based on prior knowledge, making it challenging to delineate precise boundaries of changed objects. In contrast, pixel-based change detection operates at the pixel level, identifying changes by comparing the distance between pixel pairs in feature space across dual-temporal images. As shown in Fig. 1, pixel-based change detection methods generally include three network structures: single encoder–decoder (SED), dual encoder and single decoder (DESD), and dual encoder–decoder (DED).

SED was initially designed based on semantic segmentation networks, such as fully convolutional networks (FCN) [10] and represented by UNet [11]. Dual-temporal images are fused through channelwise concatenation or convolution and directly used to generate the final change mask [12]. However, this approach merges dual-temporal features at the input stage and lacks in-depth exploration of single-temporal features, leading to suboptimal performance in change detection. DESD uses Siamese neural networks with shared weights [13] as encoders to separately extract features from dual-temporal images and then fuse them in a single decoder to generate change masks. This structure performs multiscale and deep feature extraction on the input images, but during the fusion stage, the dual-temporal features can interfere with each other, making it challenging to extract clear boundaries of the change objects. To address these limitations, Dong et al. [14] utilized pixel difference and cosine similarity during feature fusion to emphasize change features, reducing the interference between dual-temporal features to some extent. However, the introduction of pixel difference can also increase the model's sensitivity to false changes caused

Received 13 September 2024; revised 16 October 2024; accepted 31 October 2024. Date of publication 1 November 2024; date of current version 6 December 2024. This work was supported in part by the National Natural Science Foundation of China under Grant 42471457, in part by the National Key R&D Program of China under Grant 2021YFB3901305, in part by the Key Laboratory of Land Satellite Remote Sensing Application, Ministry of Natural Resources of the People's Republic of China, under Grant KLSMNR-K202206, and in part by the Postgraduate Research and Practice Innovation Program of Colleges and Universities in Jiangsu Province, under Grant KYCX24_3034. (Corresponding author: Wei Liu.)

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Digital Object Identifier 10.1109/JSTARS.2024.3490754

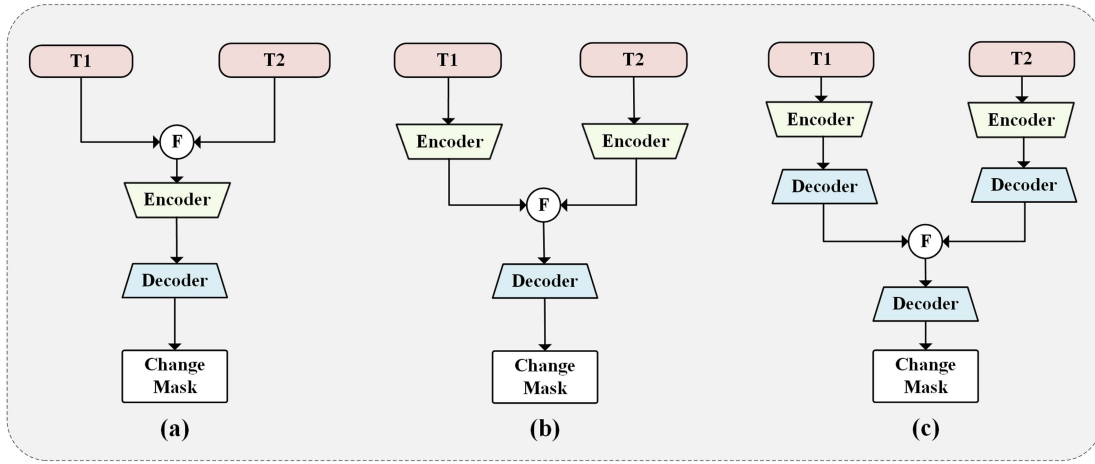


Fig. 1. Overview of three mainstream SCD architectures. (a) SED. (b) DESD. (c) DED.

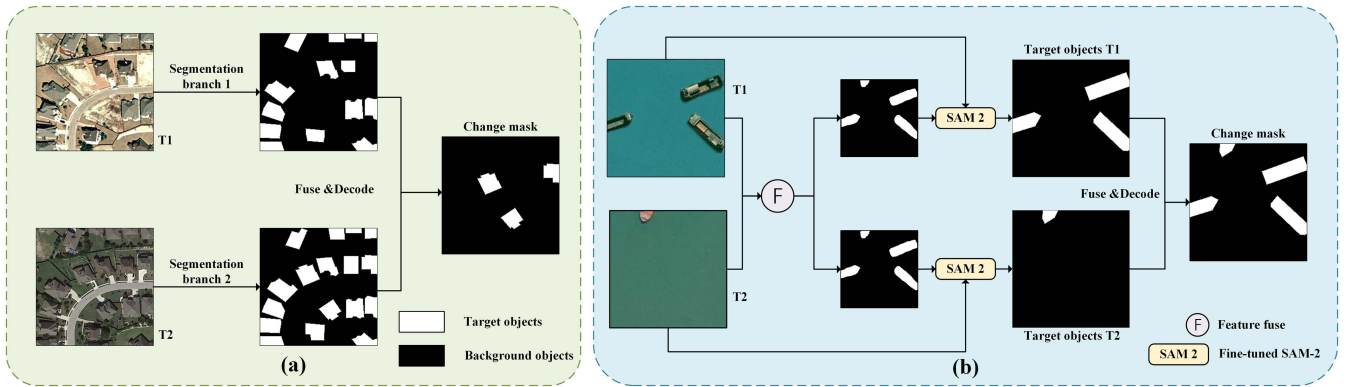


Fig. 2. Illustration of the change detection architectures. (a) DED. (b) DED-SAM.

by intraclass variability. As shown in Fig. 2(a), DED uses two encoder–decoder branches, akin to separate segmentation networks, to extract dual-temporal features and separate target features from the background for generating the final change mask. DED, on one hand, reduces the background’s impact on change mask generation, resulting in clearer boundaries. On the other hand, it leverages extracted dual-temporal features to constrain the change mask creation process, effectively aiding the model in identifying pseudochanges due to seasonal variations, spatial offsets, and other factors. Earlier DED models relied heavily on semantic labeling of bi-temporal images for training segmentation networks, significantly increasing the demand for manual annotations and leading to performance improvements that did not justify the high labor costs. Chen et al. [15], inspired by self-supervised learning [16], divided the segmentation results from the dual-branch into change regions and unchanged regions. For the unchanged regions, the segmentation result of one branch is used as the label for the other branch. In contrast, for the change regions, the opposite of the semantic segmentation resulting from one branch is used as the label for the other branch. Liang et al. [17] designed a loss function that computes the cosine similarity between dual-temporal segmentation results, aiming to minimize

differences in unchanged regions while maximizing differences in changed regions. However, these methods essentially use weakly supervision for semantic segmentation of dual-temporal images, which lacks real semantic labels and makes it difficult to accurately segment the target objects in dual-temporal images.

In recent years, vision foundation models (VFM) [18], [19] have emerged, marking a significant leap in the scale of data used for visual models. The segment anything model (SAM) [20], a large foundation model specifically designed for segmentation tasks, possesses impressive zero-shot or few-shot generalization capabilities, enabling precise boundary extraction with prompts such as points, boxes, masks, and text. Recently, Meta introduced SAM 2 [21], an advancement based on SAM, aimed at achieving rapid visual segmentation in both videos and images. SAM 2 enhances network efficiency and integrates multilevel features by using the MAE [22] pretrained Hierarchical Image Encoder [23]. It boasts an inference speed that is six times faster than SAM and achieves higher accuracy in image segmentation.

To address the challenge of accurately delineating target object boundaries in DED, we propose a novel strategy that integrates SAM 2 into DED for change detection, as shown in Fig. 2(b). Specifically, we first fuse the dual-temporal features after the encoder and input them into a dual-branch decoder,

transforming the semantic segmentation networks into small-scale change detection networks to obtain coarse change masks. Next, the coarse change masks, SAM 2 encoded features, and features from the dual-branch encoder are jointly fed into a prompt encoder to generate dense and sparse prompts, which are then used to produce dual-temporal feature maps with refined boundaries. Finally, the decision layer incorporates features from the dual-branch decoder along with the fine boundary features output by SAM 2, enabling precise identification of change objects.

Based on this strategy, we propose a DED change detection method integrating SAM 2, named DED-SAM. The main contributions of this study are as follows.

- 1) We propose a novel DED framework that transforms the semantic segmentation branches of DED into change detection branches, thereby eliminating the dependency of DED on segmentation labels.
- 2) We fine-tune the SAM 2 and integrate it with the change detection framework, effectively overcoming the limitations of traditional DED in accurately extracting dual-temporal change features.
- 3) We employ a channel exchange network (CEN) to merge dual-temporal features along the channel dimension and design a spatial fusion attention module (SFAM) to integrate spatial features.

II. RELATED WORKS

A. Deep Learning-Based Change Detection Methods

In recent years, deep learning methods, with their advanced feature extraction and complex data processing capabilities, have been widely used in various remote sensing vision tasks, including change detection. Among these, models based on convolutional neural networks (CNNs) and vision transformers (ViTs) are considered standard choices for change detection tasks.

CNNs, with their local receptive fields and convolutional operations, excel in local feature extraction. Due to their parameter-sharing characteristics in convolutional layers, CNNs also have lower computational complexity and memory consumption, making them widely used for change detection tasks. Daudt et al. [24] introduced three different FCN versions based on U-Net: FC-EF, FC-Siam-diff, and FC-Siam-conc, which were among the earliest works to apply FCNs to change detection. Li et al. [25] built upon FCN by incorporating transformers to combine multiscale features and explore the context of the images. Khankeshizadeh et al. [26] proposed the channel-spatial attention residual block and integrated it with U-Net to highlight change features. Ye et al. [27] proposed an adjacent feature cross-fusion module that aggregates complementary feature information between adjacent levels when fusing multiscale features. However, CNNs primarily focus on local features and have limitations in capturing global changes or long-range dependencies.

ViT, through its self-attention mechanism, excels at capturing global context, making it advantageous for detecting large-scale changes. Mao et al. [28] introduced a multiscale feature aggregation method based on ViT to integrate low-level

localization with high-level semantic information. Zhu et al. [29] demonstrated ViT's effectiveness in capturing large-scale changes and developed a detail capture module to compensate for its shortcomings in local modeling. Ma et al. [30] designed an edge-assisted adaptive transformer detector, where the self-adaption vision transformer block captures both local and global information, and the edge-aware decoder sharpens object boundaries. Inspired by the Swin transformer [31], Zhang et al. [32] created a U-shaped Siamese network with Swin transformer units for change detection, further validating ViT's capability to extract global spatiotemporal information. However, pure ViTs generally demand more computational resources and memory than CNNs and may not exhibit significant advantages with smaller datasets. Therefore, this study selects CNNs as the backbone network and enhances them with attention mechanisms to capture global features, addressing CNN's limitations in global modeling and long-range dependencies.

B. Vision Foundation Models and Fine-Tuning

With the substantial expansion of data and model scales, large foundational models have achieved remarkable success across various domains, including computer vision (CV) [33]. These foundational models, pretrained on large datasets, can often be quickly adapted to multiple application scenarios through fine-tuning, demonstrating performance that rivals or even surpasses that of models specialized for single tasks, both within and outside their original domains [34]. In the field of CV, VFMs can be categorized into generative models and discriminative models. Generative models, such as variational autoencoders [35], generative adversarial networks [36], and diffusion models [37], learn the data distribution of samples to generate new data similar to the training samples. Discriminative models, on the other hand, handle visual tasks such as image classification, semantic segmentation, and object detection by learning the mapping relationship between inputs and outputs. CLIP [38] and Flamingo [39] are trained on large amounts of image-text pairs collected from the web and utilize text prompts during inference, demonstrating impressive zero-shot reasoning and generalization capabilities across various downstream tasks. In addition to text prompts, some researchers are focused on training foundational models using visual prompts. SAM [20] encodes input prompts such as points, boxes, or masks to perform class-agnostic segmentation tasks at specified locations within images. As a versatile visual foundational model, SAM has shown strong cross-domain segmentation capabilities in areas such as medicine [40], video [41], and remote sensing [42].

However, these VFMs are often trained on natural images, and thus, the general knowledge they contained has limitations when addressing specific domain tasks. In complex application scenarios, the model needs to be fine-tuned with data from specific domains. Traditional global fine-tuning can lead to catastrophic forgetting of general knowledge when dealing with large base models and small specialized datasets. Parameter-efficient fine-tuning (PEFT) reduces computational resources while minimizing additional parameters and effectively adjusts

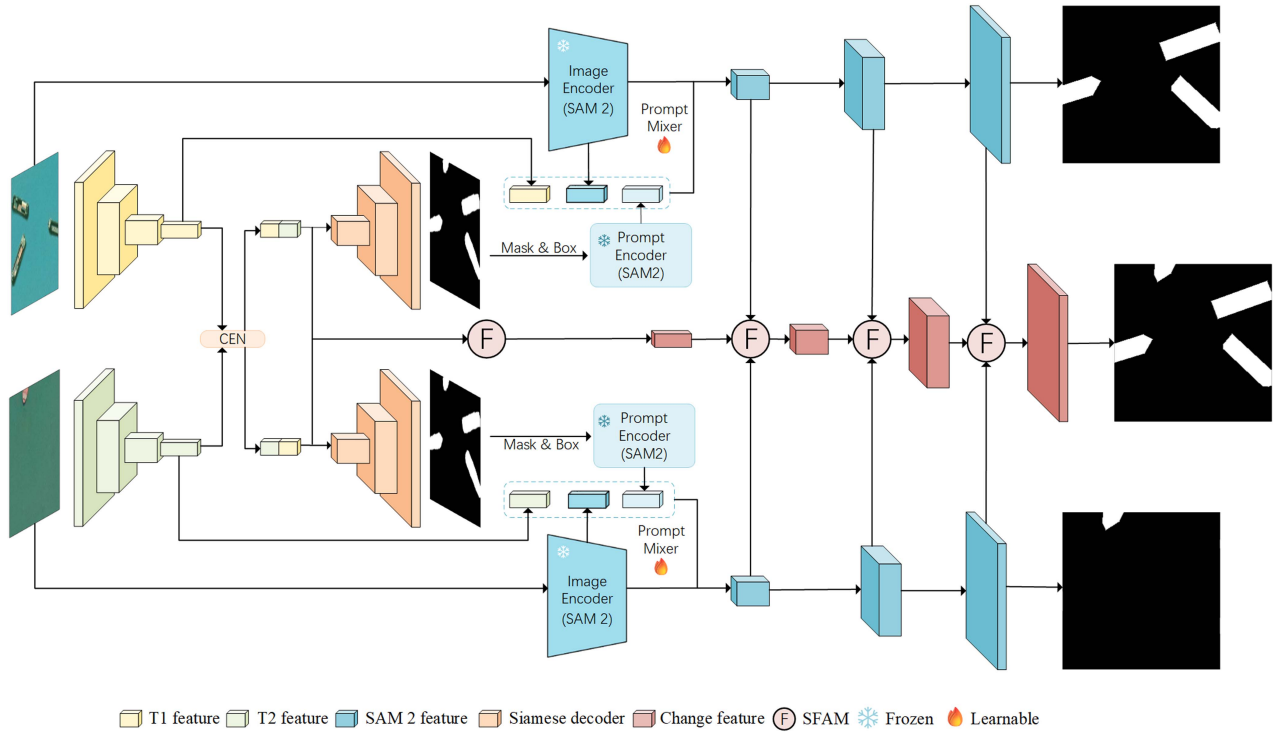


Fig. 3. Overall structure of DED-SAM.

large base models to fit specific domains [43]. PEFT is primarily categorized into three types as follows.

- 1) Adding adapters or soft prompts within the model and fine-tuning by training these additional parameters.
- 2) Fine-tuning a small subset of the model's parameters while keeping the majority frozen.
- 3) Reparameterizing model parameters using methods like LoRA, converting high-rank parameters into a combination of low-rank parameters, which significantly reduces the number of trainable parameters during fine-tuning.

Based on these methods, this study fine-tuned SAM 2 during training to better adapt it to remote sensing change detection tasks.

III. MATH

This section provides an in-depth explanation of the proposed framework. We first introduce the overall framework of DED-SAM and then detail the three core components of the framework: the DED structure, SAM 2 and its fine-tuning, and SFAM.

A. Overview of DED-SAM

As shown in Fig. 3, the proposed DED-SAM is primarily designed based on a DED structure. Unlike traditional DED, a CEN is inserted after the dual temporal twin encoders to fuse channelwise features from dual temporal phases. CEN uses scaling factors obtained from batch normalization during training to assess the importance of individual channels, achieving adaptive feature fusion. In this way, the semantic segmentation network is transformed into a small-scale change detection network. The dual temporal Siamese decoders use feature pyramid networks

to fuse hierarchical features for coarse change detection. Next, bounding boxes are generated based on the coarse change mask and input into the SAM 2 prompt encoder. These prompt embedding are combined with hierarchical features extracted by the dual temporal twin encoders and image features from the SAM 2 encoder for the decoding process. This results in the generation of smooth boundaries for temporal change objects within the dual-branch network. Finally, we designed a SFAM, which integrates the spatial features of the two branches of the SAM 2 decoder at the spatial level and ultimately generates accurate change masks.

B. DED Structure

During the encoding phase, this article employs a series of DA-ResNet modules to extract image features. As shown in Fig. 4(a), to address the limitations of ResNet [44] in global modeling and capturing long-range dependencies, we integrate the dual attention network (DANet) [45] into ResNet. DANet uses self-attention mechanisms to capture feature dependencies in both spatial and channel dimensions, thereby adaptively integrating local features with global context. Conventional DED employs small semantic segmentation networks to extract features from dual-temporal targets individually. However, the segmentation results are suboptimal due to the lack of an effective loss computation method. To address this, we introduce a CEN to adaptively fuse dual-temporal features, as shown in Fig. 4(b). Specifically, a batch normalization layer is applied after the dual-temporal encoder outputs to compute the scaling

Algorithm 1: Forward Propagation of DED-SAM.

```

// SE: Siamese Encoder
// SD: Siamese Decoder
//FP: Feature Pyramid
//SIE: SAM Image Encoder
//PE: Prompt Encoder
//SMD: SAM Mask Decoder
//PM: Prompt Mixer
//MD: Mask Decoder
Input:  $I = \{I_1, I_2\}$  (a pair of RS images)
Output:  $Z$  (a prediction change mask)
// Identify coarse change areas by dual encoder-decoder
for  $i$  in  $\{12\}$  do
     $FP_i = SE(I_i)$ 
     $Mask_{coarse}^i = SD(FP_i)$ 
end for
// Obtain precision change boundaries with SAM 2
for  $i$  in  $\{12\}$  do
     $F_i = Fusion(FP_i)$ 
     $Box_i = mask2box(Mask_{coarse}^i)$ 
     $P_i = PE(Box_i, Mask_{coarse}^i)$ 
     $F_{SAM}^i = SIE(I_i)$ 
     $FP_{SAM}^i = SMD(PM(F_i, P_i, F_{SAM}^i))$ 
end for
// Get change masks with decoder
 $Z = MD(FP_{SAM}^1, FP_{SAM}^2)$ 

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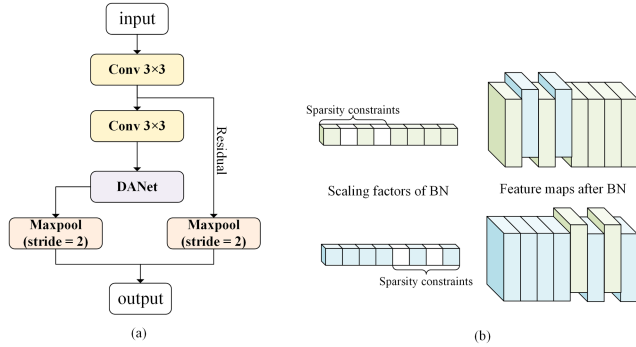


Fig. 4. Details of encoder block and CEN. Conv 3×3 denotes the convolution layer with kernel size = 3 and BN refers to the batch normal layer.

factors for each channel of the dual-temporal features

$$x'_{i,c} = \gamma_{i,c} \frac{x_{i,c} - \mu_{i,c}}{\sqrt{\sigma_{i,c}^2 + \varepsilon}} + \beta_{i,c} \quad (1)$$

where $x_{i,c}$ and $x'_{i,c}$ denote the feature maps of the i th branch in the c th channel before and after the batch normalization layer, respectively. $\gamma_{i,c}$ and $\beta_{i,c}$ denote the scaling factor and offset, respectively. $\mu_{i,c}$ and $\sigma_{i,c}$ represent the mean and standard deviation of the data within the batch. $\varepsilon > 0$ is a small parameter used to prevent the denominator from being less than or equal to zero.

When the scaling factor of a channel's feature is small after batch normalization, its impact on the final prediction is reduced and can even be deemed redundant during training. To address this, we replace channels with small scaling factors using corresponding channels from another temporal feature, effectively eliminating redundancy while fusing dual-temporal features. We use the scaling factors to measure the contribution of channel features to the prediction results and set a threshold to decide on channel swapping. Additionally, sparse constraints are applied to certain scaling factors to better identify unnecessary channel features. Notably, constraints are applied exclusively to the initial half of the T1 features and the latter half of the T2 features. This strategy is designed to mitigate redundancy in channel features within each branch and to prevent excessive channel interchanges, which could lead to substantial loss of temporal features within branches, ultimately impairing the decoder's capacity for precise change region identification. The sparse constraint is given by

$$L_s = \lambda \|\hat{\gamma}\|_1 \quad (2)$$

where L_s denotes the sparse loss, $\hat{\gamma}$ denotes the scaling factor of the selected part, and λ is a hyperparameter used to adjust the coefficient of the sparse loss in the total loss.

With CEN, each branch of DED contains dual-temporal features, enabling independent identification of change regions. Additionally, multiscale features obtained from the encoder are integrated into the decoder features via skip connections. This approach ensures that each branch includes both dual-temporal fused features and single-temporal multiscale features, facilitating precise localization of change targets and coarse extraction of change masks.

C. SAM 2 and Its Fine-Tuning

As a large-scale general-purpose vision model, the SAM 2's framework allows for various types of prompts to perform class-agnostic segmentation tasks. Specifically, SAM 2 consists of an image encoder, a prompt encoder, a mask decoder, and a memory mechanism (including memory encoder, memory bank, and memory attention module). The memory mechanism is used exclusively for video segmentation and is not discussed in this article. The image encoder is pretrained using MAE technology and the Hiera framework, the prompt encoder processes input prompts such as point boxes and masks, and the mask decoder makes predictions based on the image and prompt encodings.

Inspired by PEFT, this article designs a framework to fine-tune SAM 2 within the DED to obtain change features with clear boundaries, as illustrated in Fig. 5. Specifically, dual-temporal images are separately fed into the dual-branch Siamese encoder and SAM image encoder, producing F_{SAM}^i and F_i

$$F_{SAM}^i = SIE(I_i) \quad (3)$$

$$F_i = Fusion(FP(SE(I_i))) \quad (4)$$

where SIE and SE represent the SAM 2 image encoder and the twin encoder, respectively. i denotes the branch index, FP stands for Feature Pyramid, and $Fusion$ refers to the process of integrating multiscale features.

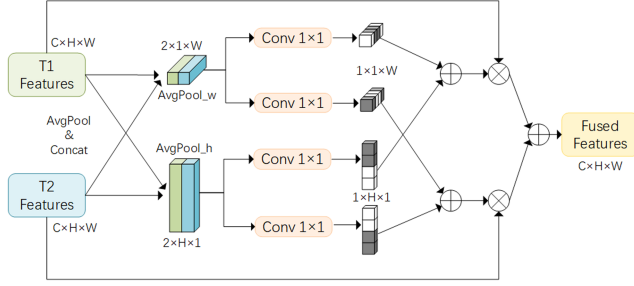


Fig. 5. Illustration of the architectures of SFAM.

Subsequently, the coarse-grained change masks generated by the Siamese decoder are used to create bounding box prompts. These prompts, along with the coarse-grained masks, are fed into the prompt encoder, resulting in the output prompt P_i

$$\text{Box}_i = \text{boundingRect}(\text{Mask}_{\text{coarse}}^i) \quad (5)$$

$$P_i = PE(\text{Box}_i, \text{Mask}_{\text{coarse}}^i) \quad (6)$$

where $\text{Mask}_{\text{rough}}^i$ and Box_i denote the coarse-grained change mask and the generated bounding box, respectively. PE denotes the prompt encoder of SAM 2.

Next, we construct a prompt mixer to integrate the semantic features extracted by the SAM 2 image encoder and the Siamese encoder, combining them with the prompts

$$P_{\text{mix}} = MLP(F_{\text{SAM}}^i, F_i, P_i). \quad (7)$$

Finally, the mixed prompts and the image features extracted by SAM 2 are input into pretrained mask decoder, producing the temporally refined change masks

$$\text{Mask}_{\text{SAM}} = \text{SMD}(F_{\text{SAM}}^i, P_{\text{mix}}). \quad (8)$$

To effectively fine-tune SAM 2 for change detection tasks, we set up a loss function that combines focal loss L_{focal} [46], Dice loss L_{dice} [47], and mean squared error loss L_{mse} , as given by the following equation:

$$L_{\text{SAM}} = L_{\text{focal}} + L_{\text{dice}} + L_{\text{mse}}. \quad (9)$$

D. Spatial Fusion Attention Module

In the context of change detection tasks, dual-temporal feature fusion is crucial. Common methods include direct fusion, convolutional fusion, and attention-based fusion. Direct fusion involves simple operations like addition, subtraction, or concatenation of dual-temporal features, which can be prone to noise and struggle with capturing complex relationships. Convolutional fusion applies convolutional operations on concatenated dual-temporal features to extract multiscale local change information, effectively reducing noise interference. Attention-based fusion uses attention mechanisms to adaptively adjust the importance of features across dimensions, followed by concatenation or addition to combine dual-temporal features. Since CEN has already implemented partial fusion of dual-temporal features in the channel dimension, we designed a SFAM to fuse spatial

information from dual-temporal features and utilize spatial attention mechanisms to identify crucial parts of these features, as shown in Fig. 5.

In SFAM, we construct two branches that perform horizontal and vertical attention on the dual-temporal features. This design allows the attention module to capture global features in one spatial direction while preserving precise positional information in the orthogonal spatial direction, thereby enhancing the model's ability to perform global modeling. In the horizontal direction, we apply global average pooling with preserved width on the dual-temporal features $F_1, F_2 \in \mathbb{R}^{c \times h \times w}$, followed by concatenation along the channel dimension. This results in $\text{Avg}_w \in \mathbb{R}^{2 \times 1 \times w}$

$$\text{Avg}_w = \text{Concat}(\text{Avg}(F_1), \text{Avg}(F_2)). \quad (10)$$

Inspired by ECA-Net [48], we designed two one-dimensional (1-D) convolutions to adaptively compute the kernel size. The kernel size k is calculated using the following formula:

$$k = \left\lceil \frac{\log_2(W)}{\gamma} + \frac{b}{\gamma} \right\rceil_{\text{odd}} \quad (11)$$

where W represents the width of the input features, b and γ are hyperparameters that adjust the ratio between width and kernel size. $\lceil t \rceil_{\text{odd}}$ denotes the nearest odd number to t .

Following this, Avg_w is fed into the two 1-D convolutions, and the outputs are processed through Softmax to ensure the values at corresponding positions sum to 1, producing the width weights W_1 and W_2 for the dual-temporal features. Similarly, the height weights H_1 and H_2 are obtained. Finally, to obtain the fused feature F_{fuse} using the dual-temporal width and height weights, the calculation is expressed as

$$F_{\text{fuse}} = (W_1 + H_1) F_1 + (W_2 + H_2) F_2. \quad (12)$$

Through SFAM, we use the attention mechanism to compute the fusion weights for dual-temporal features in both horizontal and vertical dimensions. This effectively preserves useful information in the dual-temporal features while removing redundant data. It also addresses the limitations of CNNs in global modeling and capturing long-term dependencies.

IV. EXPERIMENTS

A. Datasets

To demonstrate the superiority of DED-SAM in change detection, we conducted comprehensive experiments on three different datasets, each representing distinct application scenarios.

Single-class change detection: the LEVIR-CD [49] dataset includes 637 pairs of 0.5-m resolution remote sensing images from Google Earth, each with a size of 1024×1024 pixels. The dataset captures extensive single-class changes related to building growth and decline. In this experiment, these 637 image pairs are uniformly divided into nonoverlapping subimages of size 256×256 pixels. These subimages are then split into training, validation, and test sets, containing 7120, 1024, and 2048 image pairs, respectively.

Multiclass change detection: the SYSU-CD [50] dataset, collected and released by Sun Yat-sen University, captures

multiclass changes in Hong Kong from 2007 to 2014. The dataset includes various types of changes such as new and demolished buildings, vegetation growth and disappearance, road construction and expansion, changes in water body areas, and vessel movement. SYSU-CD consists of 20000 pairs of 0.5-m resolution aerial images, each with a size of 256×256 pixels. The dataset is divided into 12000 pairs for training, 4000 pairs for validation, and 4000 pairs for testing, adhering to a 6:2:2 distribution ratio.

Change detection under seasonal variation interference: the CDD [51] dataset is cropped from seven pairs of images with significant seasonal differences, with spatial resolutions ranging from 0.03 to 1 m. It includes multiscale change categories, from vehicles to large buildings. The dataset contains 16000 pairs of 256×256 pixel samples, divided into training, testing, and validation sets with a ratio of 10:3:3.

B. Implementation Details

The proposed framework is built on PyTorch and developed using Python 3.9. Experiments are conducted on a machine with an NVIDIA GeForce RTX 4090 running Windows 11. Various data augmentation techniques, such as random flipping, scaling, rotation, and Gaussian noise, are applied to enhance model generalization. The model is optimized using the AdamW optimizer with an initial learning rate of 0.001 and a weight decay parameter of 0.01. The number of epochs is set to 250 to ensure thorough training, and the model with the highest IoU on the validation set is saved for testing.

C. Evaluation Indicators

To evaluate the prediction results, this article uses F1 score, IoU, precision, and recall as evaluation metrics to measure the performance of the change detection method. Their calculation formulas are as follows:

$$\text{precision} = \frac{TP}{TP + FP} \quad (13)$$

$$\text{recall} = \frac{TP}{TP + FN} \quad (14)$$

$$F1 = \frac{2}{\text{recall}^{-1} + \text{precision}^{-1}} \quad (15)$$

$$\text{IoU} = \frac{TP}{TP + FN + FP} \quad (16)$$

where TP denotes the number of correctly identified positive samples, FP denotes the number of falsely reported negative samples, TN denotes the number of accurately identified negative samples, and FN denotes to the number of missed positive samples.

From the above formulas, it is evident that the F1 score provides a comprehensive assessment of both precision and recall, while IoU evaluates the overlap between predicted and true values. Therefore, in this article, we use F1 score and IoU as the primary evaluation metrics to assess the model's performance.

TABLE I
COMPARISON ON LEVIR-CD DATASETS

Model	IoU(%)	F1(%)	Rec(%)	Prec(%)
FC-EF	71.53	83.40	80.17	86.91
FC-Diff	74.47	85.37	83.31	87.53
FC-Conc	71.96	83.69	76.77	91.99
STANet	78.70	88.16	91.37	85.04
BiT	80.68	89.31	89.37	89.24
ChangeFormer	82.66	90.50	90.18	90.83
HMCNet	83.05	90.74	89.82	91.68
AMTNet	83.08	90.76	89.71	91.82
StransUNet	<u>83.67</u>	<u>91.11</u>	89.82	<u>92.43</u>
AFCF3D-Net	83.12	90.74	90.17	91.35
DED-SAM	85.11	92.00	<u>90.47</u>	93.51

TABLE II
COMPARISON ON SYSU-CD DATASETS

Model	IoU(%)	F1(%)	R (%)	P (%)
FC-EF	61.43	76.10	78.40	73.94
FC-Diff	62.71	77.08	79.60	74.72
FC-Conc	61.61	76.24	78.00	74.57
STANet	61.56	76.20	78.63	73.93
BiT	64.71	78.57	76.77	80.47
ChangeFormer	64.86	78.68	<u>79.88</u>	77.52
SNUNet	66.02	79.54	<u>75.87</u>	<u>83.58</u>
IFN	66.90	80.17	79.37	80.98
MSFCTNet	66.03	79.54	78.67	80.43
DARNet	<u>68.11</u>	<u>81.03</u>	79.11	83.04
DED-SAM	72.01	84.15	82.76	85.52

TABLE III
COMPARISON ON CDD DATASETS

Model	IoU(%)	F1(%)	R (%)	P (%)
FC-EF	46.68	64.32	55.26	75.14
FC-Diff	50.66	67.25	54.20	88.42
FC-Conc	52.69	69.02	56.10	89.66
IFN	65.40	79.08	70.18	90.56
BiT	90.83	95.19	94.88	95.51
ISNet	89.23	94.31	93.48	95.15
SNUNet	92.77	<u>96.25</u>	<u>96.20</u>	96.30
ChangeFormer	91.25	95.42	95.40	95.45
SSANet	<u>95.20</u>	94.33	96.06	99.07
DED-SAM	95.47	97.72	97.18	<u>98.26</u>

D. Model Comparison Experiment

To further evaluate the performance of our proposed method, we conduct a quantitative analysis by comparing it with several state-of-the-art methods. To clarify, we categorize these methods into two groups.

CNN-based: FC series [24], STANet [49], HMCNet [52], AFCF3D-Net [27], SNUNet [53], IFN [54], DARNet [55], ISNet [56], SSANet [57].

Transformer-based: ChangeFormer [69], BIT [59], TransUNet [25], AMTNet [60], MSFCTNet [61].

The experimental results on three datasets are presented in Tables I–III. The best and second-best performances are highlighted in bold and underlined, respectively. It is clear that our proposed DED-SAM demonstrates superior performance across all three change detection scenarios.

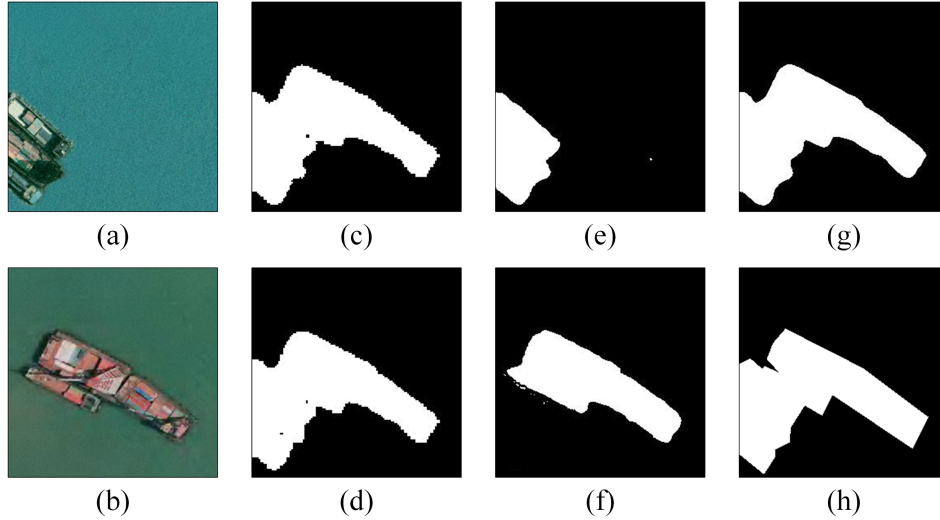


Fig. 6. Process masks and final output of DED-SAM on the SYSU-CD dataset. (a) T1. (b) T2. (c) Coarse mask 1. (d) Coarse mask 2. (e) SAM out 1. (f) SAM out 2. (g) Change mask. (h) Ground truth.

Specifically, on the LEVIR-CD dataset, DED-SAM achieved the best performance in all metrics except recall, attaining the highest IoU (85.11%) and the highest F1 score (92.00%). Compared to the second-best algorithm, StransUNet, our method shows improvements of 1.44% in IoU and 0.89% in F1 score, demonstrating the effectiveness of our approach in detecting single-class changes. Additionally, although our method has a lower boundary recall rate compared to some existing methods, such as STANet, these methods often achieve higher recall scores at the expense of significantly reduced precision, resulting in an overall F1 score that is lower than that of DED-SAM.

On the SYSU-CD dataset, DED-CD achieved even more satisfactory results. Compared to the second-best method, DARNet, DED-CD improved by 3.9% in IoU and 3.12% in F1 score, reaching optimal precisions of 72.01% and 84.15%, respectively. SYSU-CD includes change objects with multiple classes and semantic ambiguity, which poses a significant challenge for conventional change detection methods. Thanks to its dual-temporal feature extraction capabilities, DED-SAM effectively captures change objects with clear boundaries in both temporal images. By comparing these objects, it generates the final change mask, thus enhancing its performance in complex multiclass change detection scenarios.

Fig. 6 illustrates the process masks and final output of DED-SAM during inference, visually demonstrating its superiority in extracting dual-temporal change features. Specifically, the dual-branch encoder-decoder section generates coarse masks 1 and 2 based on the dual-temporal features, as shown in Fig. 6(c) and (d). Subsequently, SAM 2 uses the dual-temporal images, coarse masks, and bounding boxes generated from coarse encoding to produce the corresponding temporal change features, as depicted in Fig. 6(e) and (f). Ultimately, these dual-temporal change features are used to generate a change mask with well-defined boundaries.

DED-SAM also demonstrates impressive performance on the CDD dataset, achieving the highest IoU (95.47%) and the highest F1 score (97.72%), surpassing the second-best methods by

0.37% and 1.47%, respectively. Notably, models that performed well on LEVIR-CD, such as the FC series and BiT, exhibit a significant performance drop on the CDD dataset. This decline is primarily due to their reliance on simple concatenation or subtraction techniques for fusing dual-temporal features, which can lead to feature interference and struggle with seasonal variations that introduce false changes. In contrast, DED-SAM leverages CEN and SFAM to effectively assess the importance of dual-temporal feature channels and spatial dimensions during the fusion process. This approach significantly mitigates the impact of seasonal false changes, resulting in superior performance on the CDD dataset.

Figs. 7 and 8 present the visualization results of DED-SAM on the LEVIR-CD and CDD datasets, respectively, with false negatives marked in red and false positives in blue. The visualization results demonstrate that for small-scale changes, such as buildings and trees, DED-SAM exhibits lower false detection and missed detection rates compared to other methods. Additionally, for large-scale changes, such as roads with significant edge features, DED-SAM also shows better boundary detection accuracy.

E. Ablation Experiment

In this section, we continue using the aforementioned evaluation metrics as the basis for comparison to conduct detailed ablation experiments on DED-SAM. The goal is to validate the effectiveness of each component of our proposed method. The specific results are presented in Table IV. Where DESD stands for the dual encoder and single decoder network. DED stands for the DED structure. CEN stands for the CEN. SFAM stands for the SFAM.

The analysis of the ablation experiment results leads to a significant conclusion: DED-SAM outperforms both DESD and DED on both datasets. Furthermore, each module contributes to the performance improvements to varying degrees. It is noteworthy that DED-SAM achieves a significantly larger improvement

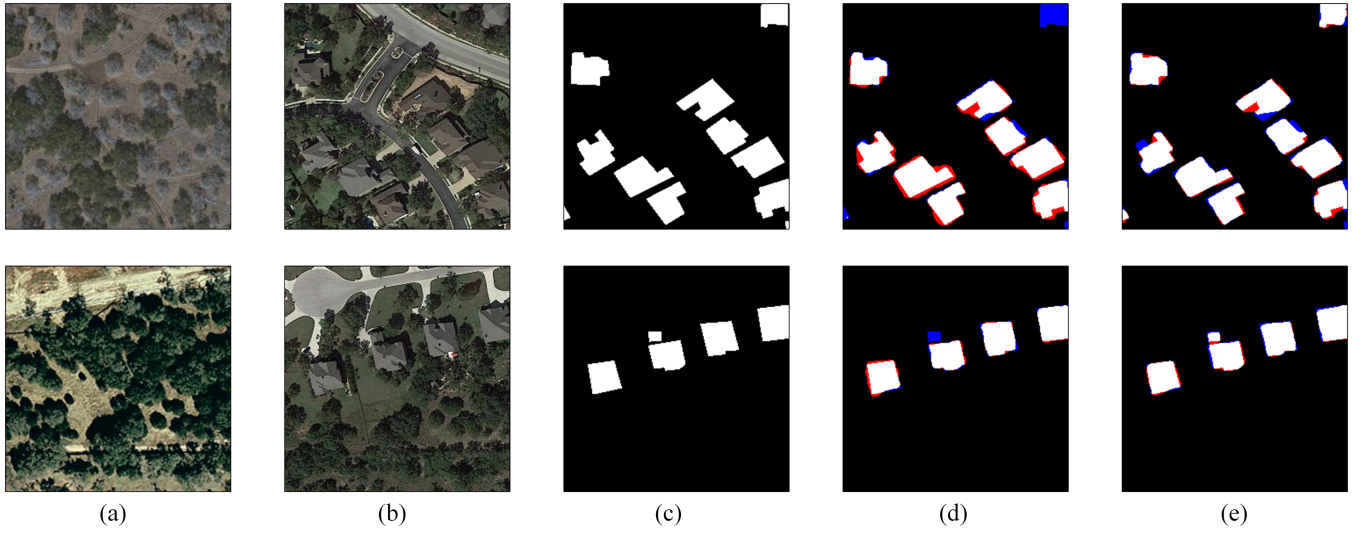


Fig. 7. Visualization results on Levir-CD. Red areas denote false positives and blue areas denote false negatives.

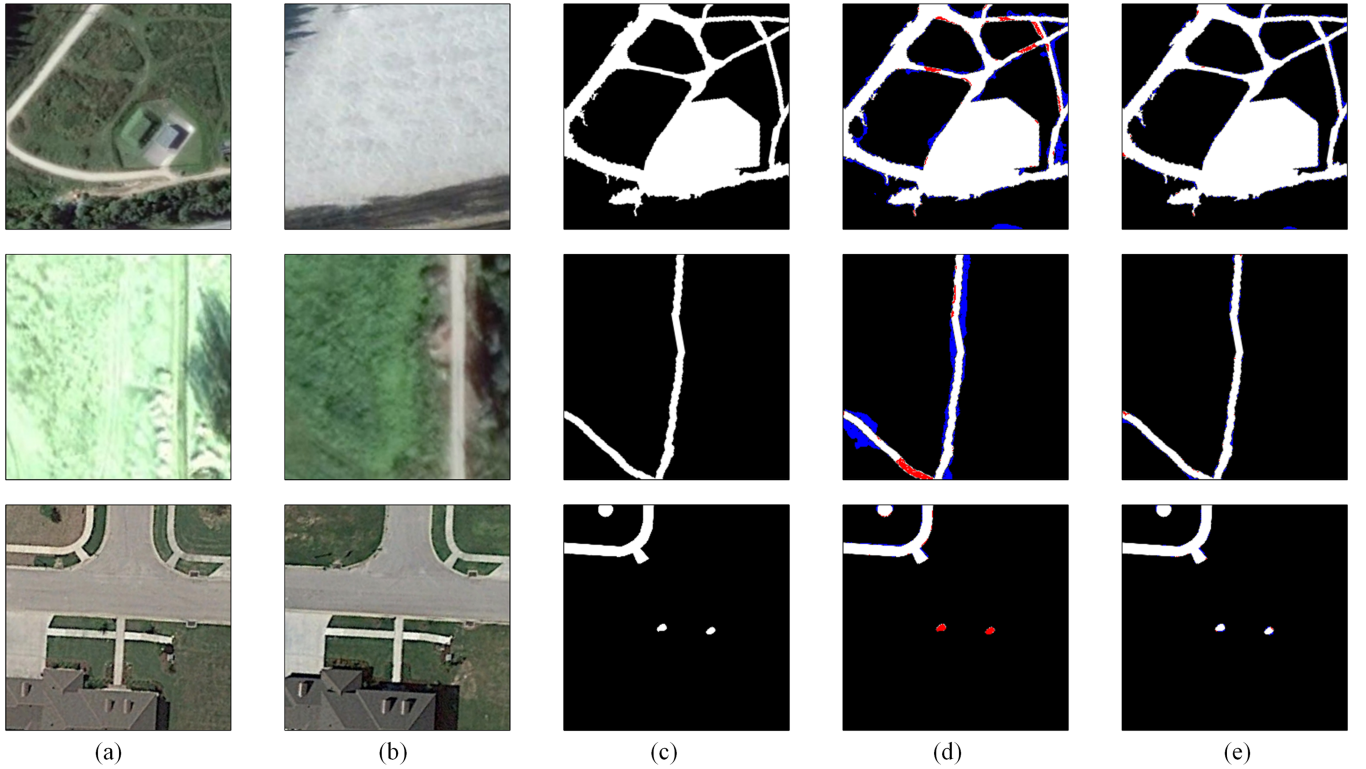


Fig. 8. Visualization results on CDD.

over DED on the SYSU-CD dataset (IoU improvement of 4.92% and F1 score improvement of 3.88%) compared to the Levir-CD dataset (IoU improvement of 1.31% and F1 score improvement of 0.81%). Based on the previously described principles of DED-SAM, we propose the following explanations for this observation.

- 1) Levir-CD contains only a single class of change with precise image registration, which is not particularly challenging even for simpler existing models. In contrast,

SYSU-CD includes multiple change classes and more complex scenes, which better leverages the capability of DED-SAM to separately extract dual-temporal change features.

- 2) In this study, we need to use coarse change masks to generate prompts for SAM 2. Buildings in Levir-CD are usually smaller in scale and higher in density, which may cause some coarse change masks to stick, thus degrading the segmentation performance of SAM 2.

TABLE IV
ABLATION STUDY OF DED-SAM ON LEVIR-CD AND CDD DATASETS

Model	Dataset	IoU(%)	F1(%)	R(%)	P(%)
DESD	Levir	82.64	90.47	90.12	90.76
DED	Levir	83.80	91.19	89.44	<u>93.01</u>
DED+CEN	Levir	84.06	91.41	90.40	92.44
DED+CEN+SFAM	Levir	<u>84.78</u>	<u>91.76</u>	90.67	92.88
DED-SAM	Levir	85.11	92.00	<u>90.47</u>	93.51
DESD	SYSU	65.82	79.18	74.37	84.89
DED	SYSU	67.09	80.27	75.69	<u>85.48</u>
DED+CEN	SYSU	69.05	81.69	78.24	85.46
DED+CEN+SFAM	SYSU	<u>71.16</u>	<u>83.27</u>	<u>81.65</u>	84.78
DED-SAM	SYSU	72.01	84.15	82.76	85.52

- 3) The attention module proposed in this study focuses on addressing CNN's deficiencies in global modeling, thereby improving the ability to capture global changes. Since Levir-CD includes changes that are often minor and localized, the SFAM's effectiveness is not very pronounced.

In summary, we posit that although DED-SAM demonstrates superior performance across a range of scenarios, its primary strength is evident in multiclass change detection tasks involving significant large-scale change.

V. DISCUSSION

The recent state-of-the-art methods for change detection often focus on optimizing submodules of the network through the addition of attention mechanisms or modifications to the backbone. Alternatively, customized modules are integrated into the model to enhance its performance for specific tasks. These methods usually involve changes to the overall structure of the model only to a limited extent. This article focuses on the framework structure of change detection and aims to address some deficiencies in the traditional DED framework by incorporating a recently popular foundational model—SAM 2. However, there is still much room for improvement in our work.

- 1) In this article, we utilize the adaptive CEN to achieve the transition from the semantic segmentation branch to the change detection branch. This parameter-free feature fusion method effectively reduces the complexity of the model. However, we believe that it is possible to explore a new fusion method that can achieve better feature fusion while not introducing excessive parameters.
- 2) In our study, we utilize the coarse-grained masks obtained from the dual-branch decoder to prompt SAM2, thereby extracting fine-grained change features from the dual-temporal images. During this process, we found that the fine-grained change feature maps generated by SAM 2 may contain not only the corresponding temporal feature boundaries but also some noise introduced by the coarse-grained masks. This suggests that our fine-tuning of SAM 2 may not be thorough. We believe that adopting a better fine-tuning method could lead to a significant improvement in the accuracy of DED-SAM.

VI. CONCLUSION

This article introduces a change detection method that integrates SAM 2 with DED, named DED-SAM. Specifically, we employ CEN to adaptively exchange dual-temporal channel features, allowing the dual-branch decoder to localize change objects and output coarse-grained change masks. Subsequently, SAM 2 is fine-tuned to extract dual-temporal change features based on these coarse-grained masks. Additionally, we design SFAM to fuse dual-temporal features spatially for the final change mask generation. To validate the efficacy of our approach, comprehensive experiments were conducted on three datasets representing different application scenarios. The results indicate that DED-SAM effectively addresses the issue of accurate dual-temporal feature extraction that existing DED networks face. It demonstrates strong performance across various application scenarios, particularly in multi-class change detection tasks. Looking ahead, we plan to further optimize the network's deficiencies and apply it to more complex real-world tasks, such as nonagriculturalization detection and illegal land use inspection.

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